PROGRAMMING LAB REFERENCE GUIDE

Arithmetic Operations Program

This reference guide provides complete implementations, step-by-step logic, code walkthroughs, and formatting guidelines for inputting two numbers and displaying their **sum, difference, product, and quotient** across major programming languages (C, C++, Python, and Java).

1. Algorithm & Logic Flow

1. **Start** the program execution.
2. **Declare variables** (`num1`, `num2`, `sum`, `difference`, `product`, `quotient`) to store inputs and calculation results.
3. **Prompt user for input:** Read the first number into `num1` and the second number into `num2`.
4. **Perform Arithmetic Calculations:**
   - `sum = num1 + num2`
   - `difference = num1 - num2`
   - `product = num1 * num2`
5. **Division Error Checking:**
   - If `num2 != 0`, compute `quotient = num1 / num2`.
   - If `num2 == 0`, display an error message indicating division by zero is undefined.
6. **Display the calculated outputs** clearly on the screen.
7. **End** the program.

2. Program Implementations

C Language Implementation

```
#include <stdio.h>

int main() {
    float num1, num2;
    float sum, difference, product, quotient;

    // Prompt user for input
    printf("Enter first number: ");
    scanf("%f", &num1);

    printf("Enter second number: ");
    scanf("%f", &num2);

    // Compute basic operations
    sum = num1 + num2;
    difference = num1 - num2;
    product = num1 * num2;

    // Display basic results
    printf("
--- Results ---
");
    printf("Sum: %.2f
", sum);
    printf("Difference: %.2f
", difference);
    printf("Product: %.2f
", product);

    // Division with zero check
    if (num2 != 0) {
        quotient = num1 / num2;
        printf("Quotient: %.2f
", quotient);
    } else {
        printf("Quotient: Division by zero is undefined!
");
    }

    return 0;
}
```

C++ Implementation

```cpp
#include <iostream>
using namespace std;

int main() {
    double num1, num2;

    cout << "Enter first number: ";
    cin >> num1;

    cout << "Enter second number: ";
    cin >> num2;

    cout << "
--- Results ---" << endl;
    cout << "Sum: " << (num1 + num2) << endl;
    cout << "Difference: " << (num1 - num2) << endl;
    cout << "Product: " << (num1 * num2) << endl;

    if (num2 != 0) {
        cout << "Quotient: " << (num1 / num2) << endl;
    } else {
        cout << "Quotient: Division by zero is undefined!" << endl;
    }

    return 0;
}
```

Python Implementation

```python
# Input two numbers from user
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

# Calculate arithmetic operations
sum_val = num1 + num2
diff_val = num1 - num2
prod_val = num1 * num2

# Display results
print("
--- Results ---")
print(f"Sum: {sum_val}")
print(f"Difference: {diff_val}")
print(f"Product: {prod_val}")

if num2 != 0:
    quot_val = num1 / num2
    print(f"Quotient: {quot_val}")
else:
    print("Quotient: Division by zero is undefined!")
```

Java Implementation

```
import java.util.Scanner;

public class ArithmeticOperations {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter first number: ");
        double num1 = scanner.nextDouble();

        System.out.print("Enter second number: ");
        double num2 = scanner.nextDouble();

        System.out.println("
--- Results ---");
        System.out.println("Sum: " + (num1 + num2));
        System.out.println("Difference: " + (num1 - num2));
        System.out.println("Product: " + (num1 * num2));

        if (num2 != 0) {
            System.out.println("Quotient: " + (num1 / num2));
        } else {
            System.out.println("Quotient: Division by zero is undefined!");
        }

        scanner.close();
    }
}
```

3. Sample Execution / Console Output

```
Enter first number: 20
Enter second number: 4

--- Results ---
Sum: 24.00
Difference: 16.00
Product: 80.00
Quotient: 5.00
```

4. Step-by-Step Code Walkthrough (C Example)

- **Header File (`#include <stdio.h>`):** Imports input/output routines such as `printf` and `scanf`.
- **Variable Setup (`float num1, num2...`):** Allocates memory for input numbers and result variables.
- **Reading Inputs (`scanf`):** Captures user inputs from the console into memory locations.
- **Basic Operations (`+`, `-`, `*`):** Performs addition, subtraction, and multiplication.
- **Conditional Check (`if (num2 != 0)`):** Ensures the denominator is non-zero before attempting division to prevent runtime errors.

- **Formatted Output (`printf`):** Displays values formatted to two decimal places (`%.2f`).

5. MS Word Formatting Instructions

1. **Insert Code Table:** Go to Insert > Table > select a 1x1 Table.
2. **Add Background Shading:** Under Table Design > Shading, choose a light grey tint (e.g., #F4F4F4).
3. **Apply Monospaced Font:** Highlight code inside the cell and set the font to Consolas or Courier New at 9.5 pt.
4. **Set Tight Spacing:** Set Line Spacing to 1.15 and Paragraph Spacing Before/After to `0 pt`.